

CSE-251

ELECTRONIC CIRCUITS

LECTURE-11

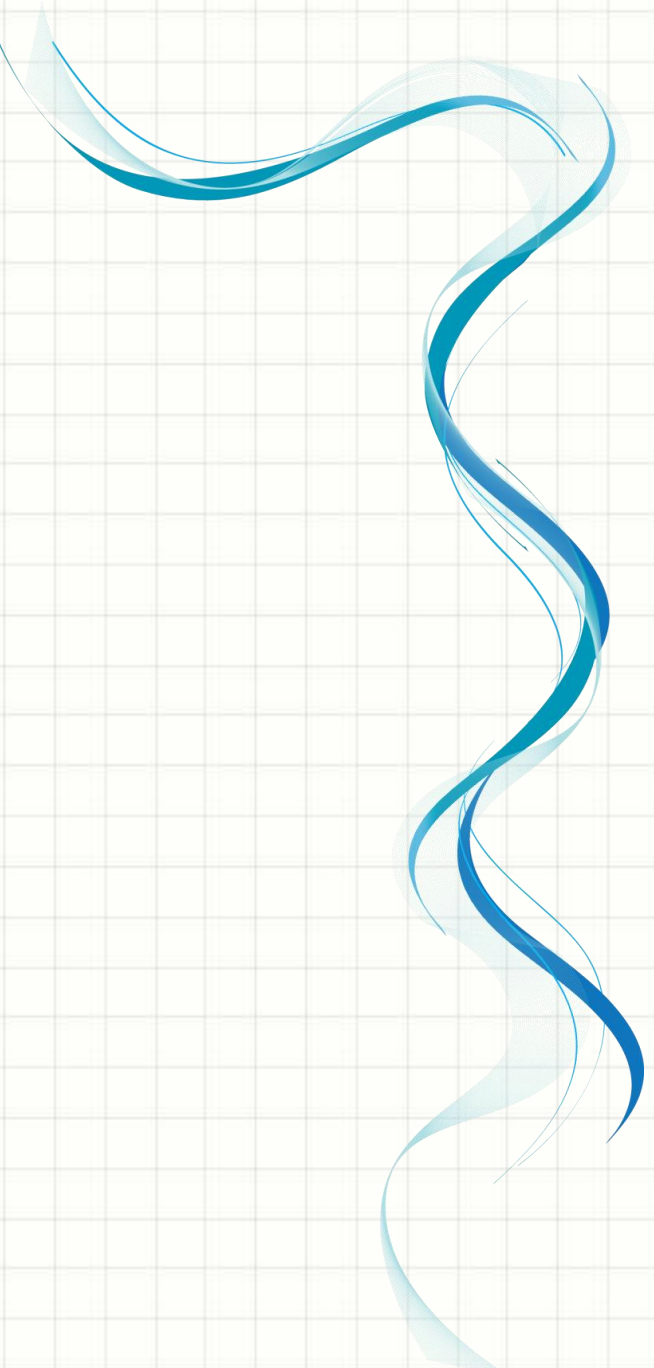
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Lecturer
Department of CSE

Today's Overview:

- ❑ *Bipolar Junction Transistor*
- ❑ *Modes of Operations*

Yo, done
with the mid?
A'ight, here
we go again!



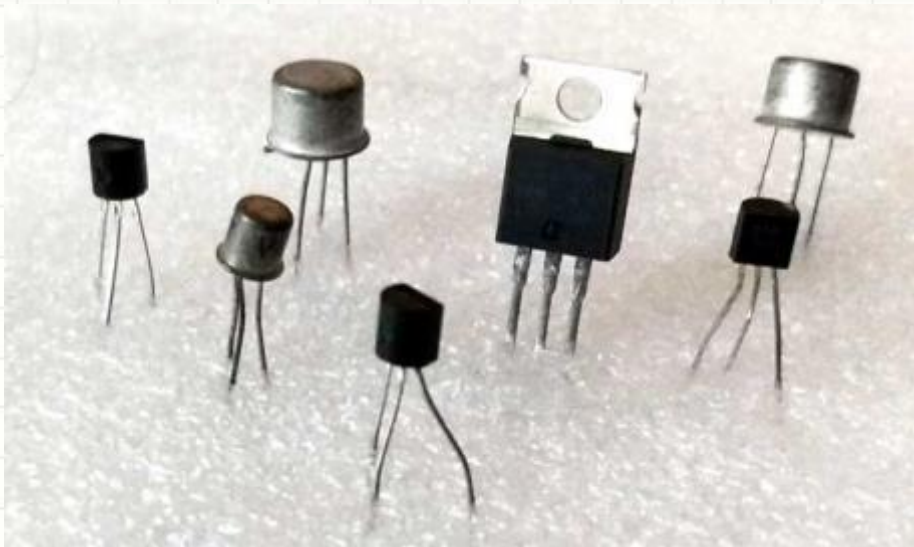
A decorative blue wavy line with a light blue shadow, flowing vertically on the left side of the slide.

Bipolar Junction Transistor

Transistor:

A “Transistor” is semiconductor device for amplifying, controlling and generating electrical signals.

Transistors are one of the key components in most of the electronic devices that are present today. Developed in the year **1947** by three American physicists- **John Bardeen**, **Walter Brattain** and **William Shockley**, the transistor is considered as one of the most important inventions in the history of science.



Dr. William Shockley (seated); Dr. John Bardeen (left); Dr. Walter H. Brattain. (Courtesy of AT&T Archives and History Center.)

Dr. Shockley Born: London, England, 1910
PhD Harvard, 1936

Dr. Bardeen Born: Madison, Wisconsin, 1908
PhD Princeton, 1936

Dr. Brattain Born: Amoy, China, 1902
PhD University of Minnesota, 1928

All shared the Nobel Prize in 1956 for this contribution.

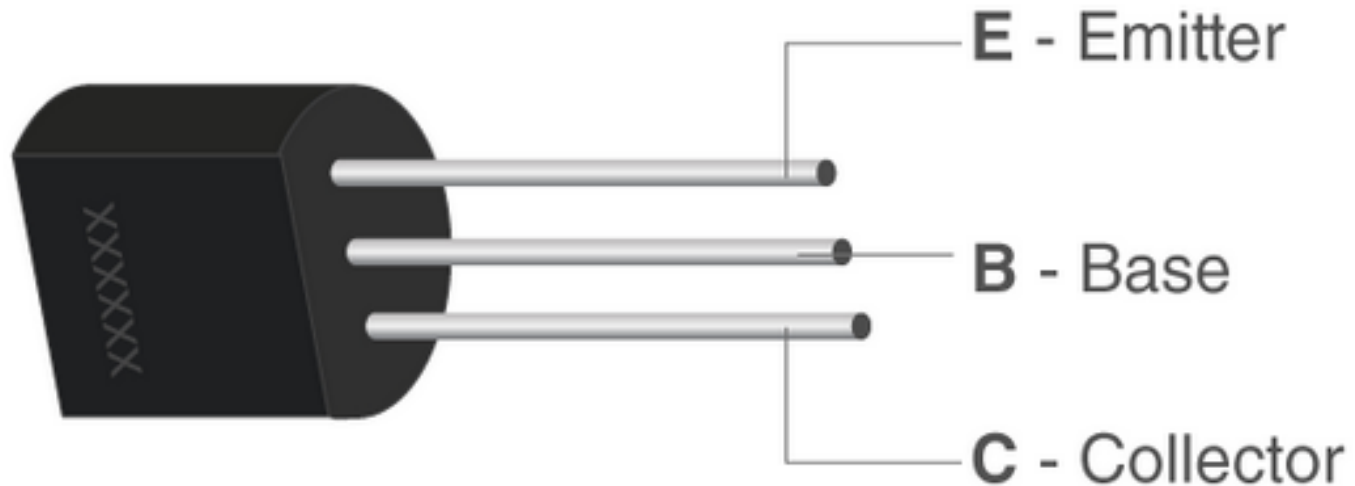
Transistor terminals:

A typical transistor is composed of three layers of semiconductor materials or more specifically, ***terminals*** which helps to make a connection to an external circuit and carry the current. They are:

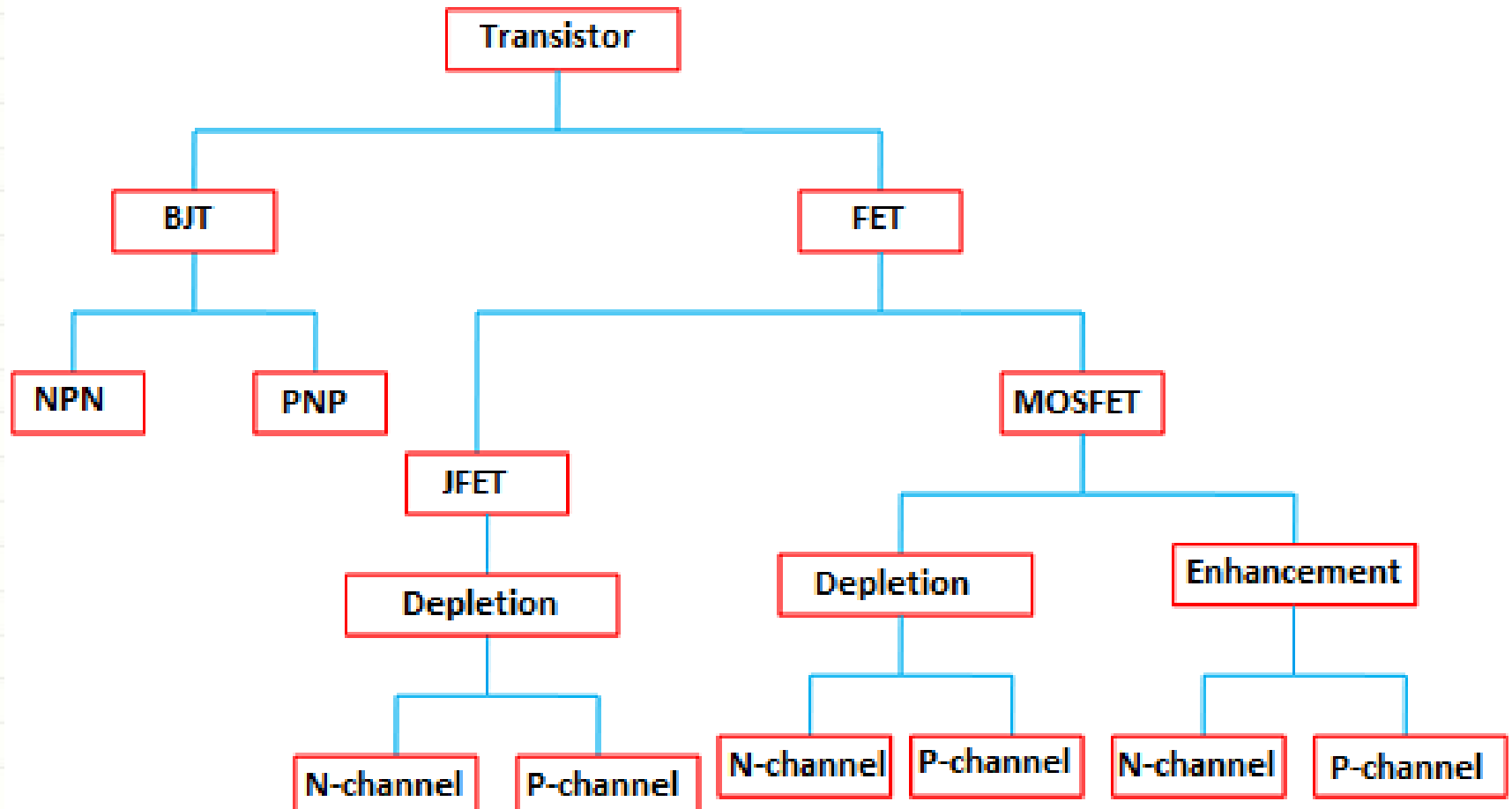
Base: This is used to activate the transistor.

Collector: It is the positive lead of the transistor.

Emitter: It is the negative lead of the transistor



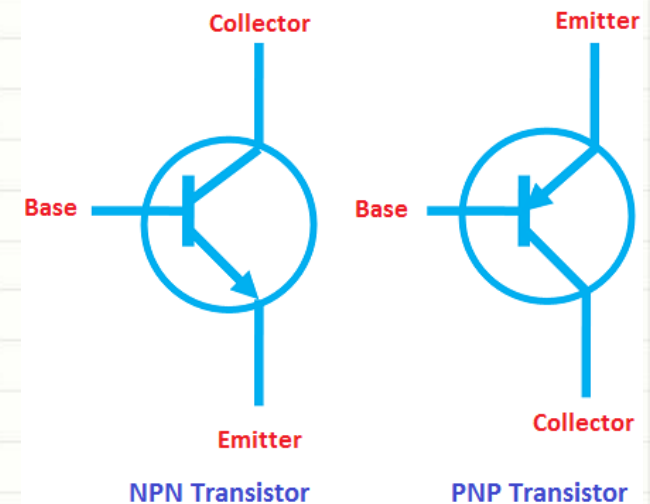
Transistor types:



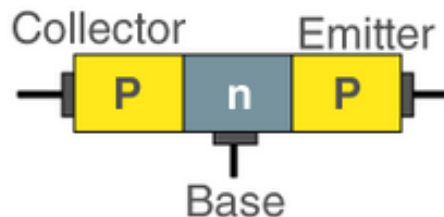
Transistor types:

A bipolar junction transistor consists of two p-n junctions formed by sandwiching either p-type or n-type semiconductor between a pair of opposite types. So there are two types of semiconductors namely:

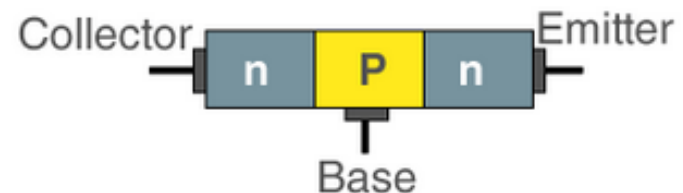
1. n-p-n transistor,
2. p-n-p transistor



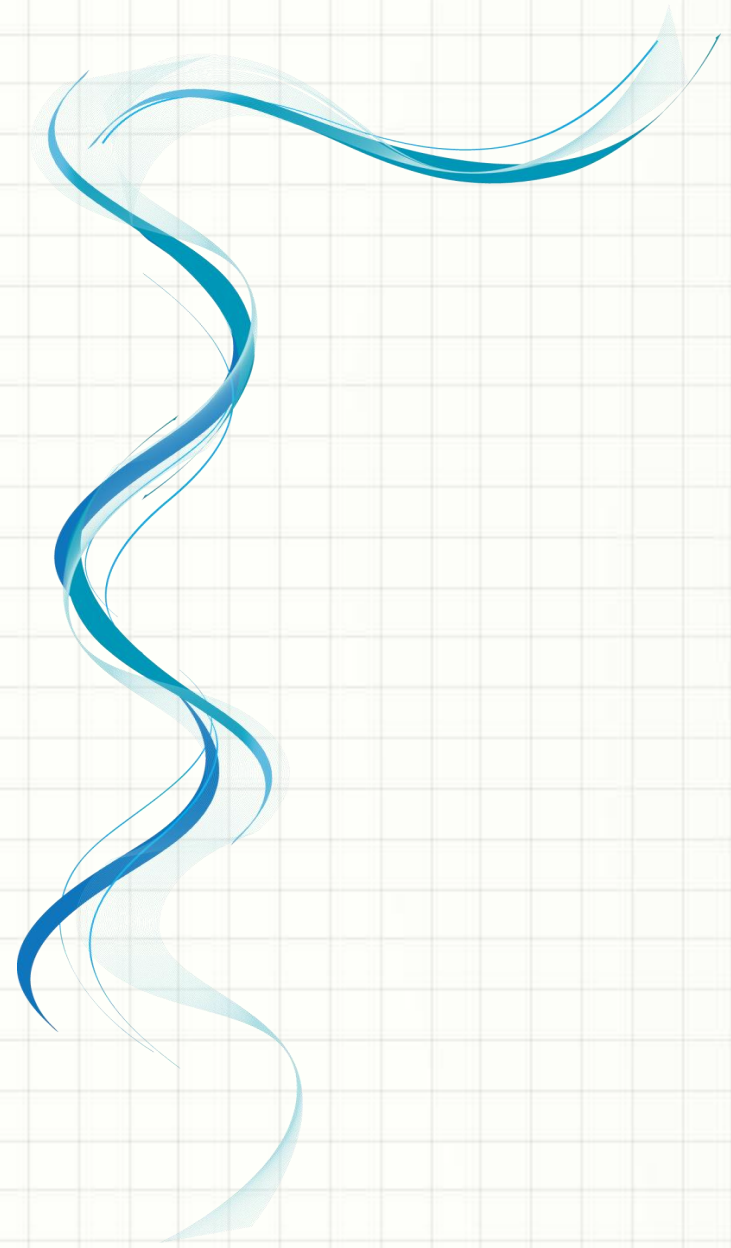
PNP



NPN



Modes of Operations



Transistor Operation Modes:

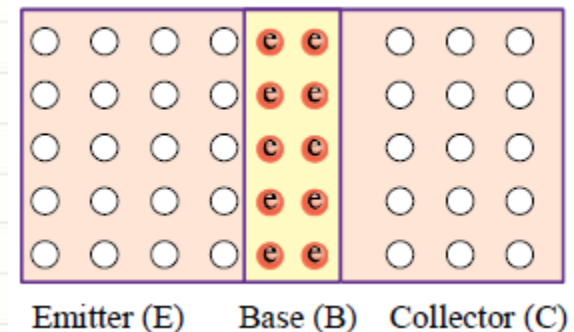
Like diodes, transistors are also biased. Biasing means applying a voltage difference in between the terminals of a transistor. The biasing of a transistor is as follows:

Base emitter bias	Base collector bias	Configuration name
Forward	Reverse	Forward active
Forward	Forward	Saturated
Reverse	Reverse	Cut off
Reverse	Forward	Reverse active

The comparison of the thickness/width and doping levels of the three regions of the transistor is as follows:

In terms of thickness: collector > emitter > base

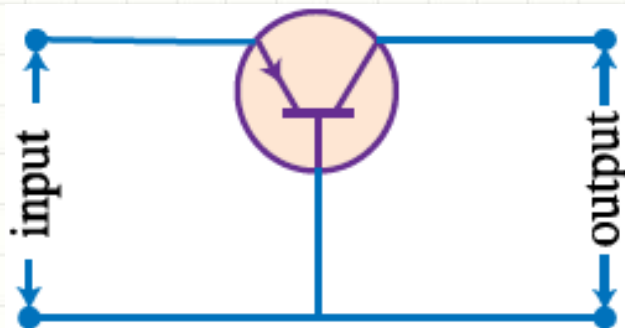
In terms of doping level: emitter > collector > base



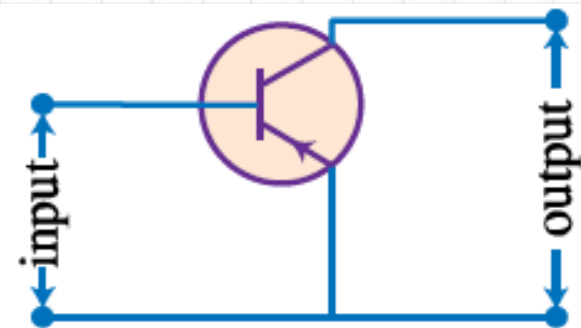
Basic configuration of Transistor:

The transistor has three types of configurations based on which terminal is in common connection. They are as follows:

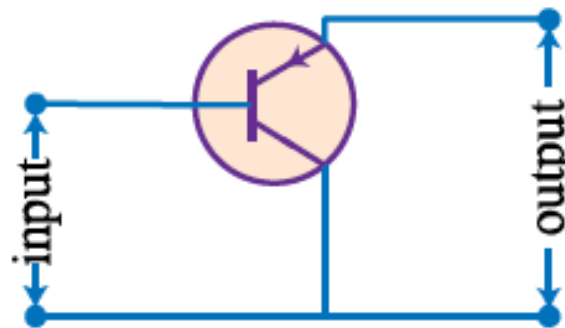
1. Common Base Configuration
2. Common Emitter Configuration (**mostly used**)
3. Common Collector Configuration (**least used**)



(a) Common Base Configuration

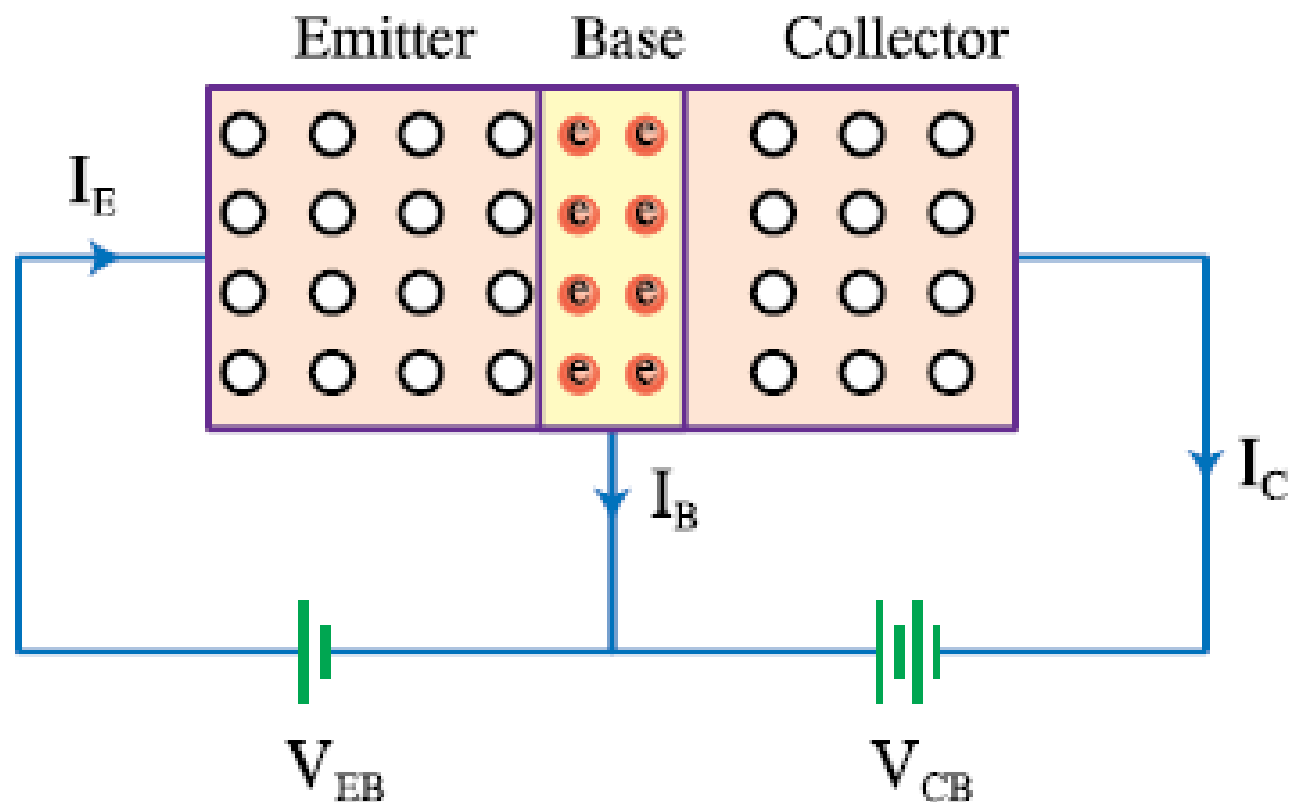


(b) Common Emitter Configuration



(c) Common Collector Configuration

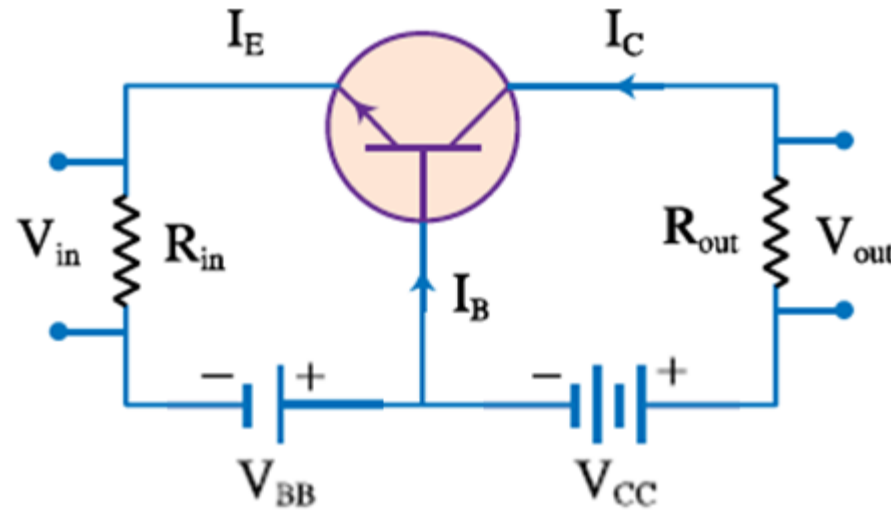
Common Base Configuration (p-n-p):



$$I_E = I_B + I_C$$

Common Base Configuration (n-p-n):

□ BJT as an amplifier:



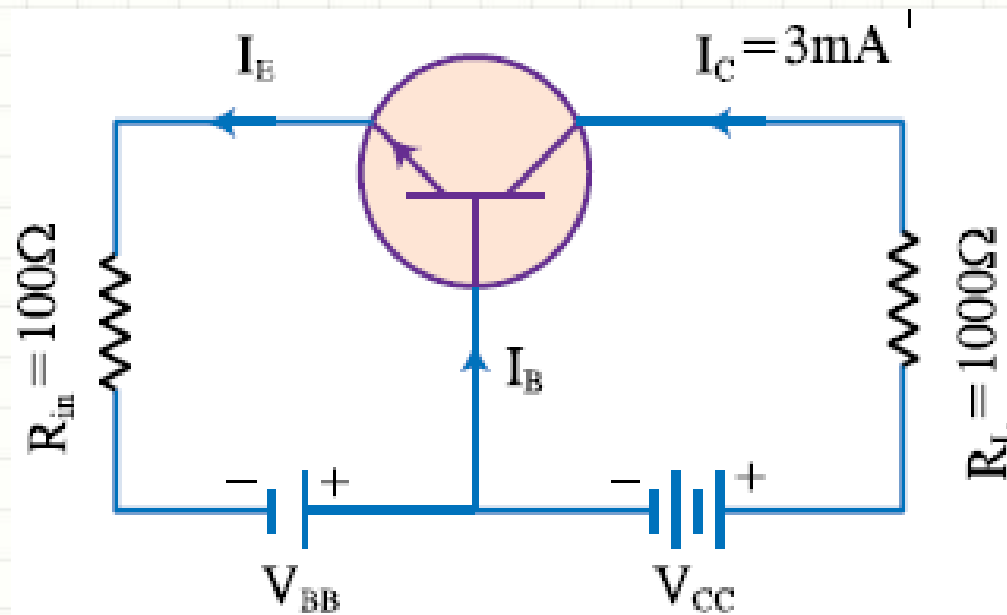
$$\text{Current Amplification Factor, } \alpha = \frac{I_C}{I_E} = \frac{\Delta I_C}{\Delta I_E}$$

$$\text{Voltage Gain} = \alpha \times \frac{R_{out}}{R_{in}}$$

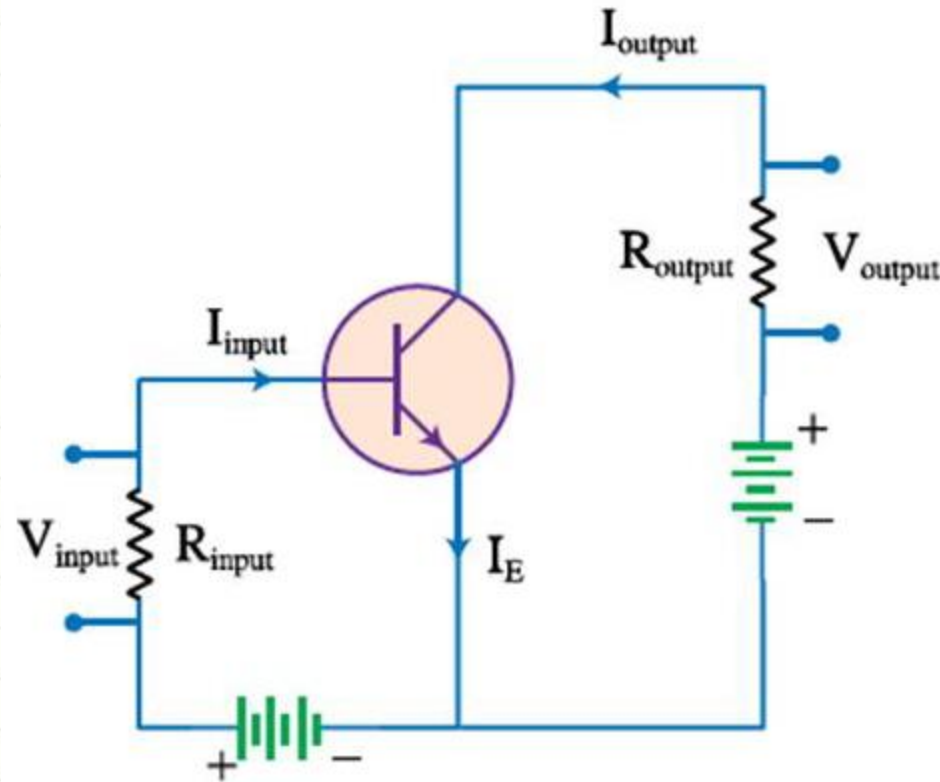
$$\text{Power Gain} = \alpha^2 \times \frac{R_{out}}{R_{in}}$$

Problem:

If the current gain is 0.99 in the mentioned circuit, determine the emitter current, voltage gain, and power gain.



Common Emitter Configuration (n-p-n):



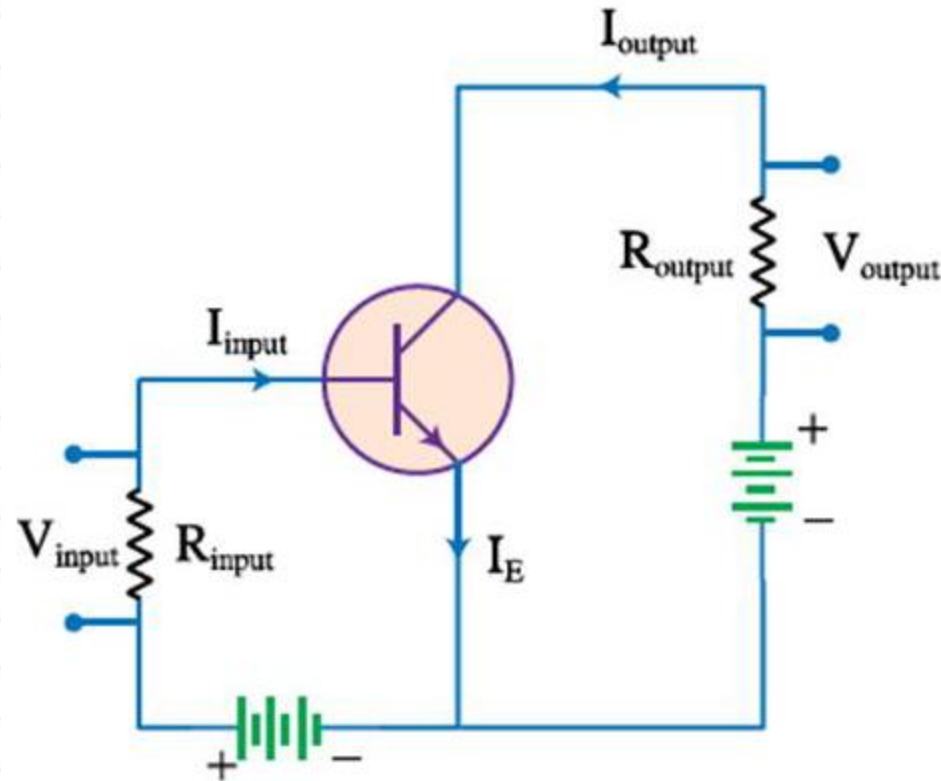
$$\text{Current Gain, } \beta = \frac{I_C}{I_B} = \frac{\Delta I_C}{\Delta I_B}$$

$$\text{Voltage Gain, } A_V = \beta \times \frac{R_{out}}{R_{in}}$$

$$\text{Power Gain, } A_P = \beta^2 \times \frac{R_{out}}{R_{in}}$$

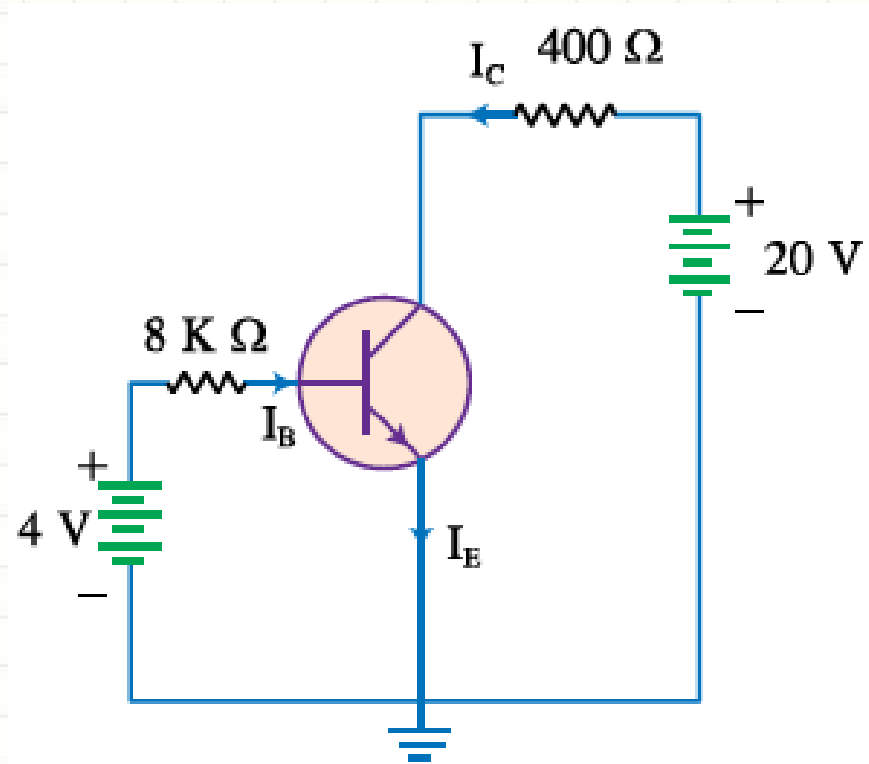
Common Emitter Configuration (n-p-n):

□ BJT as a switch:



Problem:

Find the current gain, voltage gain, and power gain for the circuit.





QUESTIONS?

SEE YOU IN THE NEXT CLASS

